

ASCII MASTER FLOW – PANEL 1/12: Cycle accounting frame

SYSTEM BOUNDARY -> atmosphere, water, soil, sediment and living biomass included

POOL -> amount held in one reservoir at a stated time

FLUX -> transfer between reservoirs per stated time

PROCESS -> biological, chemical or geological transformation

BALANCE -> inputs minus outputs; never assumed from one isolated stock

MUST REMEMBER: Biogeochemical cycles move elements among reservoirs through biological,...

ASCII MASTER FLOW – PANEL 2/12: Reservoir classification

GASEOUS -> major atmosphere or ocean reservoir

SEDIMENTARY -> major crust, soil or sediment reservoir

CARBON AND NITROGEN -> prominent atmospheric exchange

PHOSPHORUS -> rock-weathering and sediment pathway

CAUTION -> class does not set one universal cycling speed

ASCII MASTER FLOW – PANEL 3/12: Carbon movement

ATMOSPHERIC OR DISSOLVED CARBON -> photosynthetic fixation

PRODUCER BIOMASS -> consumers and detrital pathways

RESPIRATION -> carbon returned during metabolism

DECOMPOSITION AND COMBUSTION -> additional return pathways

RULE -> stored carbon and annual transfer are different measures

ASCII MASTER FLOW – PANEL 4/12: Nitrogen transformation rail

ATMOSPHERIC NITROGEN -> fixation -> ammonia or usable nitrogen forms

ORGANIC NITROGEN -> ammonification -> ammonia

AMMONIA -> nitrification -> nitrite -> nitrate

NITRATE -> plant uptake or denitrification

DENITRIFICATION -> return toward atmospheric nitrogen

ASCII MASTER FLOW – PANEL 5/12: Nitrogen organism map

RHIZOBIUM -> symbiotic fixation associated with legume roots

AZOTOBACTER -> free-living fixation example in the owner

NITROSOMONAS -> ammonia to nitrite step

NITROBACTER -> nitrite to nitrate step

PSEUDOMONAS-TYPE -> denitrification route

ASCII MASTER FLOW – PANEL 6/12: Phosphorus path

PHOSPHATE ROCK -> weathering -> soil or water phosphate

PLANT UPTAKE -> consumer transfer -> detrital return

RUNOFF -> aquatic loading -> sedimentation

MINING AND FERTILISER -> accelerate extraction and dispersal

BOUNDARY -> no significant atmospheric phase in the owner

ASCII MASTER FLOW – PANEL 7/12: Cycle-specific human pressure

CARBON -> fossil-fuel combustion and land-cover change

NITROGEN -> synthetic fertiliser and combustion-related reactive nitrogen

PHOSPHORUS -> mining, fertiliser application and runoff

OUTCOMES -> climate forcing, nutrient pollution and resource loss differ

POLICY -> source control must match the cycle and reservoir

ASCII MASTER FLOW – PANEL 8/12: Eutrophication mechanism

NUTRIENT ENRICHMENT -> increased algal or plant production

ORGANIC MATERIAL ACCUMULATES -> decomposer activity rises

OXYGEN DEMAND RISES -> dissolved oxygen may decline

ECOLOGICAL STRESS -> composition changes and mortality may occur

CLASSIFY -> natural ageing versus human-accelerated cultural eutrophication

ASCII MASTER FLOW – PANEL 9/12: Three pyramid parameters

NUMBERS -> count of organisms at each trophic level

BIOMASS -> standing living material at a stated time

ENERGY -> transfer through trophic levels over time

SHAPE -> depends on parameter and ecosystem

RULE -> only the energy pyramid is always upright

ASCII MASTER FLOW – PANEL 10/12: Aquatic biomass inversion

SMALL PHYTOPLANKTON STANDING CROP -> rapid replacement

HIGH TURNOVER -> continued production over the measurement period

CONSUMER STANDING BIOMASS -> may exceed producer biomass at one instant

ENERGY TRANSFER -> still declines upward

VERDICT -> biomass inversion is not energy inversion

CLOSE DISTINCTION: Reservoir stock is not flux, residence time is not turnover rate,...

ASCII MASTER FLOW – PANEL 11/12: Heuristic and model firewall

ENERGY LOSS -> qualitative thermodynamic direction is robust

TEN-PER-CENT RULE -> owner labels it an average heuristic

OMNIVORY -> weakens a rigid one-level assignment

SEASONALITY -> changes standing stocks and feeding relations

PYRAMID -> useful summary, not a complete food web

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name pool, process, direction, limiting...

ASCII MASTER FLOW – PANEL 12/12: PYQ and evidence boundary

ROUTED CONCEPTS -> agricultural nitrogen, rock-weathered phosphorus, fixation links

BASIC PRACTICE -> reservoir, process and organism distinctions

OFFICIAL PAPER -> demand wording only

NO KEY -> no option or answer inferred

LIVE CHECK -> no rate, pool, efficiency or status imported from a stub